

BIOINFORMATICS

WEEK 1 – DAY 5

A **MULTIPLE SEQUENCE ALIGNMENT** lines up three or more DNA or protein sequences so that equivalent positions appear in the same columns.

Imagine five students reading the same sentence.

Student 1 - THE CAT SAT ON THE MAT

Student 2 - THE CAT SAT BY THE MAT

Student 3 - THE CAT SAT ON A MAT

Student 4 - THE CAT SAT ON THE BED

Student 5 - THE CAT SAT ON THE MAT

- After aligning them, the repeated words stand out.
- The same idea applies to proteins.

Biological Example - Suppose you compare insulin proteins.

Human FVNQHLCGSHLVEALYLVCGERGFFYTPKT
Mouse FVNQHLCGSHLVEALYLVCGERGFFYTPKT
Cow FVNQHLCGSHLVEALYLVCGERGFFYTPKT
Pig FVNQHLCGSHLVEALYLVCGERGFFYTPKT
Fish FVNQHLCASHLVEALYLTGERGFFYTPKT

- Notice how most positions are identical.
- That tells us insulin is highly conserved.

Scientists use MSA to:

1. **Find conserved amino acids** - These positions often indicate functional importance.
2. **Detect mutations** - Useful for studying diseases.
3. **Build phylogenetic trees** - You'll do this tomorrow.
4. **Identify protein families** - Proteins that evolved from a common ancestor often share conserved regions.
5. **Predict protein function** - Unknown proteins can sometimes be assigned functions by comparison with known proteins.

CONSERVED REGIONS are segments of DNA, RNA, or protein sequences that remain unchanged or highly similar across different species or within a population.

- Sequences preserved throughout evolution due to natural selection.
- They typically code for essential functions. Examples include enzymatic active sites, structural cores, or binding domains.
- Mutation tolerance is very low tolerance. Alterations in these regions usually disrupt basic life functions, leading to reduced fitness or disease.

Human **ATGCGATCGA**
Mouse **ATGCGATCGA**
Cow **ATGCGATCGA**
Pig **ATGCGATCGA**
Fish **ATGCAATCAA**

Columns that remain unchanged across many species are called conserved regions.

Why? Because changing them would likely harm the protein's function. Evolution tends to preserve what is essential.

VARIABLE REGIONS are segments that show significant diversity, mutations, or divergence among different species or individual organisms.

- Sequences that accumulate mutations over time without compromising basic survival.
- They drive evolutionary adaptation, molecular diversity, and specificity.
- Mutation tolerance is exceptionally high tolerance. The surrounding framework accommodates these variations while keeping the overall molecular structure intact.

Human **ATGCGATCGA**
Mouse **ATGCGATCAA**
Cow **ATGCGATTGA**

These changes may be:

- Neutral
- Species-specific
- Adaptations
- Occasionally disease-related

CONSENSUS SEQUENCE is the calculated order of the most frequent nucleotides or amino acids found at each position in a multiple sequence alignment. It represents the most common "average" sequence among a group of related, evolutionary shared genetic or protein sequences.

Imagine every species contributes one vote for each position.

Position 1:

A
A
A
A
A

Consensus = **A**

Position 2:

G
G
G
G
T

Consensus = **G**

- The consensus sequence represents the most common residue at each position.
- It summarizes the alignment.

In a sequence alignment, **GAPS** (represented as dashes, (-)) signify evolutionary insertion or deletion events (**indels**) that occurred within the DNA, RNA, or protein sequences over time. Because structural biology cannot tell whether a piece was added to one sequence or lost from another, these blank spaces are collectively called gaps.

Example:

Human **ATGCGAATCG**
Mouse **ATGCGAATCG**
Cow **ATG-GAATCG**

The - is a gap.

CLUSTAL OMEGA is a scalable multiple sequence alignment (MSA) tool designed to align three or more nucleotide or protein sequences. It uses seeded guide trees and hidden Markov models (HMMs) to handle datasets containing hundreds of thousands of sequences quickly and accurately.

It aligns multiple DNA or protein sequences and provides:

- Aligned sequences
- Conserved positions
- Consensus information
- Downloadable alignment files

PRACTICAL EXERCISE

1. Download protein FASTA sequences for insulin from:

- Human
- Chimpanzee
- Mouse
- Cow
- Zebrafish

Use UniProt or NCBI Protein.

Save them as:

```
human_insulin.fasta  
chimp_insulin.fasta  
mouse_insulin.fasta  
cow_insulin.fasta  
zebrafish_insulin.fasta
```

2. Combine them into one file : insulin_all_species.fasta

Each sequence starts with a > header and is followed by the protein sequence.

3. Open Clustal Omega (via the EMBL-EBI web interface) and Paste all five sequences.

4. Run the alignment.